

VITA SPLENDIDA

Wearable Pulse-and-Continuous Oxygen Delivery System with Breath-Triggered Delivery, Hot-Swap Power, and Integrated Humidification

Successor to the Vita Splendida-6 (Rev. B) conceptual architecture.
Re-engineered against a 30-point patient-derived requirements checklist
and the 2024–2026 commercial portable-concentrator state of the art.

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Revision	1.0 (supersedes Vita Splendida-6 Conceptual Specification Rev. B)
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Status	Conceptual design study. Not a certified medical device.
Configurations	A: fully portable. B: tethered base station.

3.50 kg

WORN MASS, DUAL-PACK

0.5–3.0 LPM

CONTINUOUS + PULSE 1–6

6.4 h

PULSE 2 RUNTIME, HOT-SWAP

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1. Executive Overview

1.1 Purpose and Scope

The Vita Splendida is a battery-powered, body-worn oxygen concentrator that delivers both breath-triggered pulse-dose oxygen and true continuous-flow oxygen from a single 3.50 kg package. It succeeds the Vita Splendida-6 Rev. B conceptual architecture and re-engineers that architecture against two external anchors: a 30-point patient-derived requirements checklist covering performance, portability, wearability, usability, safety, and cost, and the measured 2024–2026 commercial state of the art in portable oxygen concentrators (Inogen Rove 6, CAIRE FreeStyle Comfort, Rhythm P2-E7, Philips Respironics SimplyGo, and O2 Concepts Oxlife Independence).

The Vita Splendida-6 architecture established the two-stage structure this document retains: a miniaturized pressure-swing-adsorption (PSA) generator performs oxygen enrichment, and a polyethylene-film Bellows Reservoir and Pulse-Delivery Unit (BRPD) buffers and shapes delivery. The Vita Splendida keeps that structure and corrects every element of Vita Splendida-6 that failed the patient checklist or failed physical plausibility. The five defining corrections:

- **Breath-triggered delivery replaces fixed-rate pacing.** The Vita Splendida-6 pulsed on a fixed 8–30 cycles/min metronome, which wastes oxygen between breaths and loses synchrony with the user. The Vita Splendida triggers a shaped bolus off each detected inhalation, tracks 6–40 breaths per minute, and retains the metronome only as a clinician-selected fallback.
- **True continuous flow joins pulse delivery.** The Vita Splendida delivers 0.5–3.0 LPM continuous on battery power and a 4.0 LPM High-Flow Boost on external power, serving users whose prescriptions pulse-only devices do not cover.
- **Power moves to dual hot-swappable 98 Wh packs with an internal bridge battery.** Users swap a depleted pack for a charged one mid-activity without power interruption, and each pack sits under the 100 Wh threshold that governs unrestricted air-travel carriage.
- **Physics is re-dimensioned honestly.** The Vita Splendida-6 paired a 3.5–5.0 L/min compressor intake with a claimed 6 L/min product ceiling, a combination no real PSA chain sustains at 90% purity. The Vita Splendida specifies a variable-speed twin-scroll compressor at 6–32 L/min free air delivery, LiLSX zeolite, and a pressure-equalization cycle, sized so every quoted flow figure closes on air mass balance.
- **Patient-facing completeness arrives.** Integrated membrane humidification, a Bluetooth remote fob and app, a three-tier IEC 60601-1-8 alarm suite, washable filtration, a costed bill of materials anchored to a roughly \$1,000 target price, and an explicit warranty and support program now appear as engineered subsystems rather than afterthoughts.

Regulatory status, stated up front. The Vita Splendida is a conceptual engineering design study. It is not a certified, cleared, or clinically validated medical device. Any path to human use runs through bench prototyping, verification against IEC 60601-1, IEC 60601-1-2, IEC 60601-1-8, IEC 60601-1-11, ISO 80601-2-69, ISO 18562, ISO 10993, IEC 62304, and ISO 14971, followed by formal regulatory clearance such as an FDA Class II 510(k) or Health Canada Medical Device Licence. It never substitutes for physician-directed oxygen therapy or certified equipment. Section 17 details the pathway.

1.2 What Changes from Vita Splendida-6 Rev. B

Table 1-1. Revision-level changes, Vita Splendida-6 Rev. B to Vita Splendida Rev. 1.0

Subsystem	Vita Splendida-6 Rev. B	Vita Splendida Rev. 1.0
Delivery timing	Fixed metronome pacing, 8–30 cycles/min, no breath sensing	Breath-triggered pulse with 6–40 BPM tracking; metronome retained as clinician fallback (Sec. 5)
Continuous flow	None; pulse stream described as continuous-equivalent	True continuous 0.5–3.0 LPM on battery; 4.0 LPM Boost on external power (Sec. 5.3, 5.4)
Breath sensing	Absent	Differential pressure plus thermistor flow sensing at the cannula manifold; adjustable trigger -0.05 to -0.15 cmH ₂ O; bolus inside first 250 ms of inspiration (Sec. 5.1)
Sleep operation	Not addressed	Sleep Mode: high-sensitivity trigger, 15 s no-breath fallback to 1.5 LPM continuous with advisory (Sec. 5.5)
Mouth breathing	Not addressed	Oral-Adaptive trigger logic, bolus window stretch, open-face-mask and pendant-cannula compatibility (Sec. 5.6)
Compressor	Diaphragm or scroll, 3.5–5.0 L/min intake, 35–45 W	Oil-free twin-scroll, variable speed, 6–32 L/min free air delivery, 22–150 W by demand (Sec. 4)

Sieve	Type 5A or Li-13X, ~150 mL per bed, fixed beds	LiLSX zeolite, 210 mL per bed, tool-free replaceable cartridges, pressure-equalization cycle (Sec. 4)
Bladder (BRPD)	2.5 L LDPE bladder, cam/lead-screw plate	1.2 L LDPE/EVOH bladder sized to bolus buffering; faster response, lower mass (Sec. 6)
Battery	Single fixed 400 Wh LiFePO ₄ , 2.7–3.1 kg	2 × 98 Wh hot-swap Li-ion NMC packs + 40 Wh internal bridge, 1.07 kg total (Sec. 10)
Runtime	6 h at 63 W design draw	8.2 h at pulse 1; 6.4 h at pulse 2; 2.6 h at 2 LPM continuous, dual-pack (Sec. 10.2)
Charging	USB-C PD 65–100 W; 4.5–6 h recharge	USB-C PD 3.1 140 W; ~1.8 h dual-pack recharge; 100 W car DC adapter runs and charges (Sec. 10.3)
Worn mass	4.6–5.4 kg	3.50 kg dual-pack; 3.04 kg single-pack (Sec. 12)
Acoustics	55–65 dBA at 1 m, unenclosed	40 dBA at pulse 2, 46 dBA at pulse 6, 48 dBA at 2 LPM continuous; ducted acoustic plenum (Sec. 11)
Humidification	Optional external bubble bottle note	Integrated passive membrane passover module; HME cassette option; bubble-bottle support on Boost (Sec. 7.2)
User interface	OLED plus rate and flow dials	Sunlight-readable transfective LCD, Bluetooth remote fob, iOS/Android app, voice prompts (Sec. 8)
Alarms	Five conditions, single tier	Full IEC 60601-1-8 three-priority suite, 15 conditions, independent safety-monitor MCU (Sec. 9)
Cost and warranty	Not addressed	Costed BOM ≈ \$507 landed ≈ \$527; \$995 target price; 3-year warranty with advance exchange (Sec. 15, 16)

1.3 Compliance with the 30-Point Patient Requirements Checklist

The Vita Splendida program treats the 30-point patient-derived checklist as its contract. Each item below maps to the engineered answer and the section that implements it. Every item closes; none is deferred.

Table 1-2. Thirty-point requirements compliance matrix

#	Requirement	Vita Splendida answer	Section
1	Deliver prescribed flow at rest, activity, and sleep	Breath-triggered pulse 1–6 with activity-adaptive bolus scaling, IMU-assisted activity detection, and a dedicated Sleep Mode with no-breath continuous fallback	5.1, 5.2, 5.5

2	Provide continuous flow as an option	True continuous mode, 0.5–3.0 LPM in 0.5 LPM steps, available on battery; matches Oxlife Independence continuous capability at 41% of its mass	5.3
3	Keep up with rapid breathing	Trigger tracks 6–40 BPM with a bolus inside the first 250 ms of every detected breath; adaptive scaling preserves minute volume up to the 1,500 mL/min cap; high-rate advisory beyond 40 BPM sustained	5.7
4	Support mouth breathing effectively	Dual pressure-plus-thermal sensing discriminates oral from nasal flow; Oral-Adaptive mode stretches the bolus window; open-face-mask and pendant-cannula interfaces supported on continuous mode	5.6
5	Deliver high flow for higher needs (>4 LPM)	Pulse setting 6 delivers 86 mL boluses (1,500 mL/min ceiling) for ambulatory high-need users; High-Flow Boost delivers 4.0 LPM continuous at 90% $\pm$ 3% on external power; Config B base station delivers 5.0 LPM	5.2, 5.4, 13
6	Maintain 90% $\pm$ 3% purity for device lifespan	LiLSX sieve with ultrasonic O ₂ concentration sensor in closed loop; automatic purity derate and alarm before output falls below 82%; sieve-cartridge life indicator and 12-month replacement program hold purity across life	4, 9, 15
7	Operate quietly (<55–65 dBA)	40 dBA at pulse 2, 46 dBA at pulse 6, 48 dBA at 2 LPM continuous, 52 dBA at 3 LPM or Boost; scroll compressor plus ducted acoustic plenum, versus 55–65 dBA for Vita Splendida-6	11
8	Humidify delivered oxygen, especially at higher flows	Integrated passive membrane passover humidifier (60 mL reservoir) active on pulse $\geq$ 3 and all continuous settings; HME cassette option for pulse; bubble-bottle interface on Boost mode	7.2
9	Weigh less than 10 lb (4.5 kg), ideally less	3.50 kg (7.7 lb) with dual packs; 3.04 kg (6.7 lb) with a single pack; mass budget closes item-by-item in Section 12	12
10	6+ hours battery with quick recharge	8.2 h at pulse 1 and 6.4 h at pulse 2 on dual packs; 140 W USB-C PD recharge in $\sim$ 1.8 h; 80% charge in $\sim$ 55 min	10.2, 10.3
11	Hot-swappable batteries without power-down	Dual-bay architecture with ideal-diode OR-ing and a 40 Wh internal bridge battery sustains full operation through a swap of unlimited duration	10.1
12	Built-in battery as standard	Internal 40 Wh bridge battery is non-removable and standard; it carries 1.2 h at pulse 2 alone and backs every swap	10.1
13	Easy to carry, multiple options	Backpack, cross-body, shoulder-bag, and belt-clip (single-pack) carry systems; body-conforming curved shell; 2.1 m cannula reach	12
14	Charge in a car	100 W DC adapter (11–16 V input) runs the device and charges packs simultaneously at settings up to pulse 4 or 2 LPM continuous	10.3
15	Add no exertion when worn	Center of mass sits low and within 4 cm of the spine in backpack mode; curved shell eliminates bounce; 3.50 kg total sits 1.0 kg under the checklist ceiling and 1.0 kg under SimplyGo	12
16	Skin-safe biocompatible materials	All skin-contact materials (strap padding, shell touch points) pass ISO 10993-1/-5/-10 evaluation; gas pathway passes ISO 18562; latex-free throughout	12, 17
17	Resist sweat, moisture, temperature	IP22 enclosure, conformal-coated PCBAs, expanded-PTFE membrane vents, operating envelope 5–40 °C and 15–90% RH non-condensing	12

18	Durable for everyday wear	1 m drop survival per IEC 60601-1-11 mechanical regime; 15,000 h compressor design life; 8-year device service-life target	15
19	Easy, intuitive interface	Three-button control set, plain-language transfective display, voice prompts in six languages, and a two-minute warm-up; mirrors the three-button simplicity patients rate highest in the FreeStyle Comfort class	8.2
20	Adjust settings while worn	Bluetooth remote fob adjusts flow, mode, and alarm state from hand or pocket; app provides the same on a phone; on-device panel faces forward in abdominal carry	8.3
21	Remote control	Dedicated four-button BLE fob (flow up, flow down, mode, alarm silence) with haptic confirmation and 12-month coin cell; app mirrors all functions	8.3
22	Clear, easy-to-read status indicators	High-contrast transfective LCD readable in direct sun, tricolor LED ring visible at 5 m, battery percent per pack, flow and mode numerals at 18 pt glyph height	8.2
23	Easy to clean and maintain	Rinse-and-reuse intake pre-filter (weekly), tool-free sieve cartridges, dishwasher-safe humidifier reservoir, wipe-down IP22 shell; all consumables priced in Section 15	15
24	Robust alarm system	15 alarm conditions across IEC 60601-1-8 high/medium/low priorities; 65 dBA minimum audible at 1 m; visual, voice, and haptic (fob) annunciation; alarm log retained 30 days	9
25	Safe pressure-limited system (5 cmH ₂ O)	Open, unsealed delivery interface operates at 2–5 cmH ₂ O; spring pop-off vents at 5 cmH ₂ O; the design excludes sealed positive-pressure ventilation	7.3
26	Dual flow limiters (electronic + mechanical)	Electronic proportional valve under closed-loop control plus a fixed calibrated orifice that hard-caps gas-path flow at 6 LPM independent of all electronics	7.3
27	Pressure relief on downstream occlusion	Spring-loaded pop-off at 5 cmH ₂ O vents to atmosphere on kinked tubing; occlusion alarm follows within 3 breaths	7.3, 9
28	One-way valves prevent backflow	Silicone duckbill valves at BRPD inlet and outlet block rebreathing of exhaled gas and backflow into the PSA reservoir	6
29	Reliable, with solid warranty	3-year device and compressor warranty (5-year option), 2-year sieve cartridges, 1-year batteries and accessories; 48-hour advance-exchange program; 24/7 support line; 30-day return	15
30	Affordable, target ≈ \$1,000	Costed BOM \$507 plus \$20 warranty reserve at 10,000 units/year; \$995 direct-channel target price; comparator devices list at \$2,795–\$3,995	16

2. Product Positioning and the External Lung-Assist Landscape

2.1 Design Intent and Naming

Name: Vita Splendida, from the Latin *vita* (life) and *splendida* (splendid, radiant): the splendid life. The name states the design intent in two words: oxygen therapy exists to hand the user a splendid life back, through mobility, quiet, and all-day autonomy. The device

carries a seven-day-per-week, 24-hour continuous-duty rating, and every design decision traces back to one or more of the thirty patient-derived requirements documented in Section 1.3.

Design goals, in priority order:

- Serve the full outpatient oxygen population from a single worn device: pulse-dose users, continuous-flow users to 3 LPM, and high-need users to 4–5 LPM through Boost mode and Configuration B.
- Track the user's breathing rather than the clock, across rest, exertion, hyperventilation, mouth breathing, and sleep.
- Keep worn mass under the 4.5 kg checklist ceiling with a 1.0 kg margin, without surrendering dual-mode capability.
- Deliver all-day mobility through hot-swappable sub-100 Wh battery packs rather than a single heavy fixed pack.
- Engineer to a roughly \$1,000 target price with a costed, volume-realistic bill of materials.
- State every physical limit honestly, in the document, where a buyer or clinician reads it.

2.2 Competitive Benchmark, 2024–2026

Table 2-1 positions the Vita Splendida against the five commercial devices that define the current state of the art. Data derives from manufacturer specification sheets, user manuals, and retail listings current to mid-2026 (see Reference Data Sources). The Vita Splendida's positioning logic: match the light pulse-only class on acoustics and interface, match the heavy dual-mode class on continuous flow, and undercut both on mass-adjusted capability and price.

Table 2-1. Vita Splendida against the 2024–2026 commercial field

Device	Mass as worn	Modes	Max pulse output	Continuous max	Runtime at setting 2	Noise at setting 2	List price	Warranty
Inogen Rove 6	2.2 kg	Pulse 1–6	1,260 mL/min	None	5.0 h (std battery)	39 dBA	\$2,795–3,295	3 yr device, 1 yr sieve
CAIRE FreeStyle Comfort	2.3 kg	Pulse 1–5	1,050 mL/min	None	4.25 h (8-cell)	43 dBA	≈\$2,500	3 yr incl. sieve
Rhythm P2-E7	1.98 kg	Pulse 1–7	1,400 mL/min	None	5.2 h	39 dBA	≈\$3,995	5 yr device
Philips SimplyGo	4.5 kg	Pulse 1–6 + continuous	2,000 mL/min	2.0 LPM	3.3 h pulse; 0.9 h at 2 LPM	43 dBA	≈\$2,500	2–3 yr by region
Oxlife Independence	≈8.6 kg incl. 2 batteries	Pulse 0.5–6 + continuous	1,920 mL/min (96 mL)	3.0 LPM	5.75 h pulse; 2.5 h at 2 LPM	40 dBA	\$2,895–3,195	5 yr device
Vita Splendida	3.50 kg (3.04 single)	Pulse 1–6 + continuous + Boost	1,500 mL/min (86 mL)	3.0 LPM; 4.0 LPM Boost on ext. power	6.4 h pulse; 2.6 h at 2 LPM	40 dBA	\$995–1,295 target	3 yr device, 2 yr sieve

Reading the benchmark honestly. The Vita Splendida does not beat the pulse-only class on mass: a Rove 6 at 2.2 kg carries 1.3 kg less than the Vita Splendida. That mass buys continuous flow to 3 LPM, a humidifier, dual hot-swap power, and Boost capability. A buyer whose prescription is pulse-only at setting 4 or below rationally chooses the lighter device, and Section 18 says so in plain terms. The Vita Splendida wins where the prescription includes continuous flow, high pulse output, humidification, or all-day hot-swap operation, and on price across the board.

2.3 External Lung-Assist Devices: Gas-Side and Blood-Side

The phrase "external lung-assist device" spans two fundamentally different engineering categories, and a credible specification states which one it occupies.

Blood-side (extracorporeal) assist drains venous blood, oxygenates it and strips CO₂ across a hollow-fiber membrane, and returns it, typically through a dual-lumen cannula. This category remains research-stage and hospital-bound. Notable programs: the Modular Extracorporeal Lung Assist System (MoELAS, University of Pittsburgh and Carnegie Mellon), which sustained one month of continuous support in an ovine model with a wearable centrifugal-pump plus membrane bundle; the Paracorporeal Ambulatory Assist Lung (PAAL, University of Pittsburgh, \$3.4M NIH-funded), targeting up to three months of ambulatory bridge-to-transplant support; the University of Maryland wearable artificial pump-lung (30-day ovine studies); and the Hemovent Mobybox (7-day ovine studies). These devices perform true gas exchange on blood and address respiratory failure that no gas-side device touches. They involve anticoagulation, vascular cannulation, and clinical staffing, and none is commercial.

Gas-side (non-invasive) assist enriches the oxygen fraction of inspired air and meters its delivery. This is the portable oxygen concentrator category, and it is the category the Vita Splendida occupies. The Vita Splendida moves no blood, exchanges no CO₂, and performs no alveolar function. Its BRPD bellows shapes delivery of PSA-enriched gas; it is a metering reservoir, not a gas-exchange organ, exactly as the Vita Splendida-6 Rev. B engineering note established. Within that honest boundary, the Vita Splendida maximizes what gas-side engineering delivers today: the highest continuous flow per kilogram in its class, breath-synchronous dosing, humidification, and all-day swappable power.

3. System Architecture

The Vita Splendida organizes into eight subsystems. Gas flows left to right through the chain; control, power, and thermal management wrap around all stages. The architecture preserves the Vita Splendida-6 two-stage insight: the PSA stage enriches, the BRPD stage buffers and shapes, and neither stage attempts the other's job.

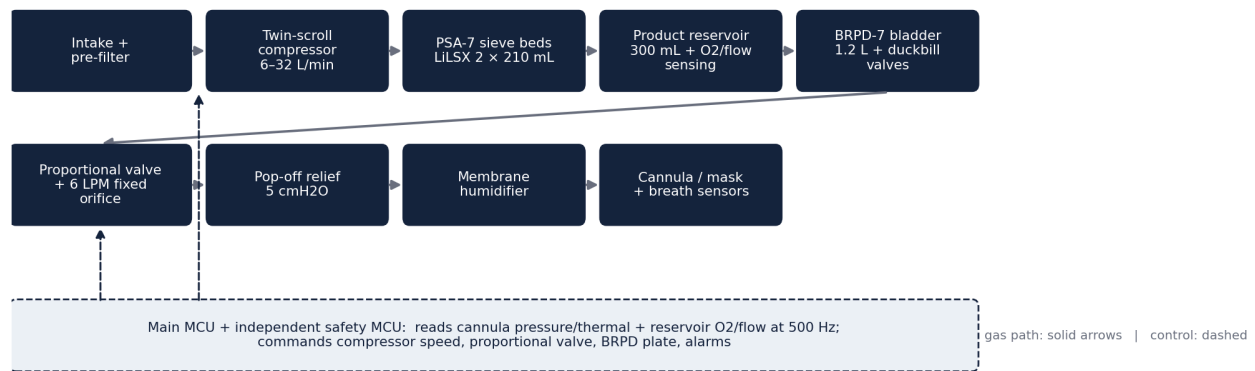


Figure 3-1. Functional architecture: gas path (solid) and control loops (dashed).

Table 3-1. Subsystem map

Stage	Subsystem	Function
1	Air intake and filtration	Draws ambient air through a washable foam pre-filter and a tool-free fine particulate filter; expanded-PTFE membrane vents block liquid water while passing air
2	Twin-scroll compressor	Oil-free, brushless, variable-speed compression to 2.0–2.5 bar gauge; 6–32 L/min free air delivery on demand (Section 4)
3	PSA-7 sieve module	Two 210 mL LiLSX zeolite beds in alternating adsorb/purge cycles with pressure equalization; produces 90% ±3% oxygen-enriched gas (Section 4)
4	Product reservoir and sensing	300 mL accumulator smooths PSA pulsation; ultrasonic O ₂ concentration sensor and thermal mass-flow sensor feed closed-loop control (Sections 4, 8)
5	BRPD-7 module	1.2 L PE-film bladder and motorized compression plate buffer enriched gas and shape boluses at high settings; duckbill valves enforce one-way flow (Section 6)
6	Flow limiting and delivery	Electronic proportional valve plus fixed calibrated orifice (6 LPM hard ceiling); 5 cmH ₂ O pop-off; membrane humidifier; cannula or mask interface (Section 7)
7	Control, sensing, and interface	Main MCU plus independent safety MCU; breath sensors; display, remote fob, app; IEC 60601-1-8 alarm system (Sections 8, 9)
8	Power	Two hot-swap 98 Wh packs, 40 Wh internal bridge battery, USB-C PD 3.1 input, car DC input, per-pack battery management (Section 10)

Gas path summary: ambient air → pre-filter → twin-scroll compressor → PSA sieve beds (nitrogen adsorption) → product reservoir (purity and flow sensing) → BRPD-7 bladder (bolus

buffering and shaping) → proportional valve → fixed orifice ceiling → humidifier → cannula or mask. **Control loop summary:** the main MCU reads cannula pressure and temperature 500 times per second, detects inspiratory onset within 40 ms, meters the proportional valve to deliver the set bolus inside the first 250 ms of inspiration, and modulates compressor speed so reservoir pressure holds 120–140 kPa gauge across demand. The independent safety MCU monitors delivered concentration, outlet pressure, temperature, and battery state, and it overrides the main MCU into a safe state on any excursion outside Section 9 limits.

4. Oxygen Generation Subsystem (PSA-7)

The PSA-7 performs the actual oxygen enrichment. It scales the same pressure-swing adsorption principle used in every commercial POC, dimensioned so that each quoted product flow closes on air mass balance. The governing physics: at 90% product purity with lithium low-silica-X (LiLSX) zeolite, a well-designed two-bed cycle with pressure equalization delivers one volume of product oxygen for roughly 8.5 volumes of compressed ambient air. LiLSX's nitrogen-over-oxygen adsorption selectivity of approximately 7 at one atmosphere (versus roughly 3 for legacy 13X) is the single largest efficiency lever in the subsystem, and published comparisons show it doubling oxygen recovery against Na-X sieve under identical cycle conditions.

Table 4-1. PSA-7 oxygen generation subsystem

Parameter	Specification
Technology	Two-bed pressure-swing adsorption with pressure-equalization step; four-step cycle (pressurize/adsorb, equalize, purge/desorb, equalize-return)
Sorbent	LiLSX (lithium low-silica-X) zeolite beads, 0.4–0.8 mm; N ₂ /O ₂ selectivity ≈7 at 1 atm; bulk density ≈0.66 kg/L; ≈280 g total charge
Sieve bed volume	210 mL per bed, two beds, in tool-free replaceable cartridges with keyed bayonet fittings and integral dust filters
Cycle timing	5–8 s adsorb per bed, ≈1 s equalization, 3–5 s countercurrent purge; timing varies with compressor speed and altitude
Compressor	Oil-free twin-scroll, brushless DC, variable speed 1,200–3,600 rpm; dynamically balanced, valve-less; Air Squared P07/P11-class lineage scaled to twin air-end
Discharge pressure	2.0–2.5 bar gauge (200–250 kPa gauge), closed-loop regulated
Free air delivery	6–32 L/min at intake, variable with demand; 8.5:1 air-to-product ratio at 90% purity anchors the sizing: 32 L/min air sustains 3.0 LPM continuous product with 15% transient margin
Product concentration	90% ±3% at all settings per ISO 80601-2-69 convention; 93% nominal at settings ≤2 LPM continuous

Product flow capacity	Pulse 1–6 to 1,500 mL/min oxygen output; continuous 0.5–3.0 LPM; Boost 4.0 LPM on external power (compressor at 150 W ceiling)
Compressor electrical draw	22 W at pulse 2 average; 48 W at 1 LPM continuous; 70 W at 2 LPM; 98 W at 3 LPM; 150 W ceiling in Boost
Module mass	1.14 kg (compressor 0.62 kg, sieve cartridges 0.36 kg, manifold and valves 0.16 kg)
Warm-up	≤2 min to specification purity at 20 °C; ≤4 min at 5 °C
Altitude compensation	Compressor speed and cycle timing auto-compensate 0–3,048 m (70–106 kPa); operation to 4,000 m with reduced-purity advisory below 87%

Product reservoir and sensing. A 300 mL rigid accumulator sits downstream of the beds and smooths the pulsed PSA output. An ultrasonic oxygen-concentration sensor (service life >5 years, replacing the 12–18 month galvanic cells of earlier designs) and a thermal mass-flow sensor mount at the reservoir outlet. The control loop trims compressor speed and cycle timing against these readings: concentration droop of 2 percentage points triggers speed-up; a further droop to 87% raises the medium-priority low-concentration advisory; droop toward 82% raises the high-priority alarm before the 82% threshold, as ISO 80601-2-69 mandates.

Why the pressure-equalization step matters. Before a bed switches from adsorb to purge, the Vita Splendida briefly connects the two beds and transfers the pressurized gas in the finishing bed into the starting bed rather than venting it. This recovers roughly 20–30% of compression work per cycle and is standard practice in the current commercial generation; it is the mechanism that lets a 22 W average draw at pulse 2 coexist with a 1,500 mL/min ceiling at pulse 6.

Plausibility note, carried forward from the Vita Splendida-6 tradition. The Vita Splendida-6 paired a 3.5–5.0 L/min intake with a 6 LPM product ceiling. At an 8.5:1 air ratio, 6 LPM of 90% product demands roughly 51 L/min of compressed air and 150–200 W of sustained compressor power, which no 40 W wearable compressor supplies. The Vita Splendida corrects the dimensioning: 32 L/min intake supports the honest ceilings of 3.0 LPM continuous on battery, 4.0 LPM Boost on external power, and 1,500 mL/min pulse output. Every flow figure in this document closes on that mass balance.

5. Breath Sensing and Delivery Modes

5.1 Breath-Trigger Subsystem

The trigger subsystem replaces Vita Splendida-6's clock with the user's own breathing. A differential piezoresistive pressure sensor and a micro-thermistor flow element mount in the cannula manifold at the device outlet, sampling at 500 Hz. The pressure channel detects the negative deflection at inspiratory onset; the thermal channel discriminates gas actually

moving past the nares from ambient pressure noise, which is what keeps the trigger honest during mouth breathing and in windy outdoor conditions.

Table 5-1. Breath-trigger subsystem

Parameter	Specification
Trigger threshold	−0.08 cmH ₂ O standard; adjustable −0.05 to −0.15 cmH ₂ O in five steps (Sleep, Sensitive, Standard, Exercise, Insensitive); reference class performance sits at 0.12 cmH ₂ O, so the Vita Splendida range brackets both weaker and stronger inspiratory efforts
Detection latency	<40 ms from inspiratory onset to trigger assertion
Bolus timing	Valve opens within 60 ms of trigger; 80% of bolus volume delivers inside the first 250 ms of inspiration, the window in which early-inspiratory oxygen reaches the gas-exchange regions most effectively
Rate tracking	6–40 BPM with per-breath triggering; no breath stacking; one bolus maximum per detected inspiration
False-trigger rejection	Dual-channel coincidence (pressure dip plus thermal flow) plus adaptive baseline tracking; rejection of speech, cough, and wind artifacts within 2 breaths
Failure fallback	Loss of trigger signal for 15 s in pulse mode raises the no-breath advisory and shifts delivery to Metronome Mode at the last tracked rate (Section 5.8); in Sleep Mode it shifts to 1.5 LPM continuous (Section 5.5)

5.2 Pulse-Dose Delivery

Pulse mode meters a discrete bolus per breath. Bolus volumes follow the industry convention of quoting at a 20 BPM reference rate, with adaptive scaling as rate changes: faster breathing draws smaller, more frequent boluses; slower breathing draws larger, less frequent boluses, holding delivered minute volume constant per setting. This mirrors the Intelligent Delivery and autoSAT behaviors of the commercial class leaders.

Table 5-2. Pulse-dose settings (20 BPM reference; tolerance ±15% or 4 mL, whichever is greater)

Setting	Bolus volume at 20 BPM	Minute volume at 20 BPM	Bolus window	Typical clinical fit
1	15 mL	300 mL/min	250 ms	Mild ambulatory need, conserve-and-carry days
2	26 mL	520 mL/min	300 ms	Standard ambulatory prescription
3	38 mL	760 mL/min	350 ms	Moderate need with exertion headroom
4	52 mL	1,040 mL/min	400 ms	High-need ambulatory

5	68 mL	1,360 mL/min	450 ms	High-need ambulatory, exertion
6	86 mL	1,500 mL/min (ceiling)	500 ms	Maximum pulse output; covers many users prescribed >4 LPM-equivalent ambulatory dosing

The 1,500 mL/min oxygen-output ceiling sits above the Rove 6 (1,260) and P2-E7 (1,400) and below the Oxlife Independence (1,920). Above 20 BPM the controller scales bolus volume downward to hold the ceiling, so a user at 40 BPM on setting 6 receives roughly 37 mL per breath rather than a doubled minute volume. Pulse settings are not liter-equivalents; clinicians titrate pulse settings to saturation by oximetry, and the device displays this convention in the user manual in the standard industry wording.

5.3 Continuous-Flow Delivery

Continuous mode delivers a steady metered stream at 0.5, 1.0, 1.5, 2.0, 2.5, or 3.0 LPM, selectable in 0.5 LPM steps. The proportional valve closes the loop on the outlet mass-flow sensor, holding $\pm 10\%$ or ± 0.1 LPM accuracy (whichever is greater) across breath-induced back-pressure fluctuation. On battery, 3.0 LPM runs 1.8 h on dual packs; 2.0 LPM runs 2.6 h; 1.0 LPM runs 3.8 h (Section 10.2). On AC or car DC power, continuous mode runs indefinitely at any setting. Continuous mode serves the populations pulse dosing misses: sleep users who trigger unreliably, mouth breathers, users on mask interfaces, and prescriptions written in liters per minute rather than pulse numbers.

5.4 High-Flow Boost Mode

Boost mode answers the checklist's >4 LPM requirement with honest physics. With the device on AC mains or the 100 W-plus car adapter, the twin-scroll runs to its 150 W ceiling and delivers 3.5 or 4.0 LPM continuous at $90\% \pm 3\%$. On battery, Boost operates for excursions up to 30 minutes with a runtime warning at activation and automatic step-down to 3.0 LPM at 20% remaining energy. Sustained 4.0 LPM continuous at 90% purity demands roughly 34 L/min of compressed air and 150 W of electrical power; a battery pack sized for that draw in wearable mass delivers minutes, not hours, and Section 18 discusses the trade rather than hiding it. Users who need true 5 LPM continuous step to Configuration B (Section 13), where the base station's larger compressor and mains power make 5.0 LPM a standard, not an exception.

5.5 Sleep Mode

Sleep Mode configures three behaviors simultaneously, following the pattern Philips proved in the SimplyGo: trigger sensitivity shifts to -0.05 cmH₂O to catch shallow nocturnal breaths; bolus windows stretch to deliver at lower amplitude over a longer interval; and the no-breath watchdog arms at 15 seconds. If no breath triggers within that window, the device shifts automatically to 1.5 LPM continuous and raises a low-priority advisory, so an open-mouth sleeper or a slipped cannula never silently zeroes delivered oxygen. Return of a

detected breath pattern restores pulse delivery. A sustained no-flow condition at the outlet (kink, disconnect) raises the high-priority alarm regardless of mode.

5.6 Mouth-Breathing Accommodation

Mouth breathing degrades every pulse-dose device, because a nasal cannula samples only nasal flow. The Vita Splendida attacks the problem on four fronts rather than pretending one sensor solves it. First, the thermal flow channel detects gas movement even when the pressure signature of a mouth-breathing cycle weakens, and Oral-Adaptive logic widens the acceptance window and stretches the bolus across the first 40% of inspiration. Second, the Vita Splendida supports an open-face-mask interface (OxyMask-class) on continuous settings 0.5–3.0 LPM, which captures oral and nasal flow together; the mask is the clinically sound answer for committed mouth breathers, and the Vita Splendida treats it as a first-class interface rather than an improvised hack. Third, a pendant-conserving cannula (Oxymizer-class) mates to the standard outlet barb for users who prefer a reservoir cannula. Fourth, Sleep Mode's continuous fallback (Section 5.5) covers nocturnal mouth breathing automatically. The honest limit stands in Section 18: no pulse-dose device fully recovers conservation efficiency in a pure mouth breather, and continuous or mask delivery is the correct prescription answer.

5.7 High Breath-Rate Behavior

The trigger tracks every detected breath from 6 to 40 BPM, including hyperventilation during exertion or anxiety. The controller never skips a detected breath and never doubles a bolus: it scales bolus volume down as rate rises, holding minute volume at the setting's ceiling. At sustained rates above 40 BPM for 60 seconds, the device raises the medium-priority high-breath-rate advisory (the same alarm class the SimplyGo publishes) and continues triggering to a 50 BPM hardware maximum, at which point delivery shifts to Metronome Mode at 40 cycles/min with the advisory maintained. The PSA stage's 15% transient air margin absorbs rate spikes without concentration droop; the reservoir level sensor confirms charge before each bolus at settings 4–6.

5.8 Metronome Fallback Mode

A clinician-selectable Metronome Mode preserves the Vita Splendida-6's fixed-rate behavior for the minority of users who prefer paced delivery or whose trigger signature is unreliable: adjustable 6–40 cycles/min, default inspiratory-to-expiratory shaping of 1:2, with the same bolus table as Section 5.2. It also engages automatically on trigger-signal loss (Section 5.1). The mode displays a persistent on-screen indicator, because paced delivery without breath synchrony conserves measurably less oxygen, and the user deserves to see which regime is active.

6. Reservoir and Pulse-Delivery Module (BRPD-7)

The BRPD-7 is the component most identified with the Vita Splendida line: a polyethylene-film bladder driven by a motorized compression plate in a breathing-like rhythm. The Vita Splendida-6 Rev. B engineering note stated its true function, and the Vita Splendida repeats it verbatim in substance: the BRPD is a pulse-dose gas reservoir, not a gas-exchange organ. It enriches nothing. It buffers PSA product gas, paces bolus delivery, and shapes high-setting boluses so the PSA stage never has to track breathing in real time. The Vita Splendida shrinks the bladder from 2.5 L to 1.2 L, because a breath-triggered architecture sizes its buffer against the largest single bolus (86 mL), not against several seconds of fixed-rate tidal volume. The smaller bladder responds faster, weighs less, and still holds fourteen times the maximum bolus.

Table 6-1. BRPD-7 reservoir and pulse-delivery module

Parameter	Specification
Bladder material	LDPE/EVOH coextruded film, 175 μm ; the EVOH core layer cuts oxygen permeation loss through the film wall to <1% per hour of residence
Bladder construction	Heat-sealed perimeter seams, reinforced inlet and outlet port fittings; bladder is a user-replaceable service item (annual, or on puncture)
Bladder volume	1.2 L nominal; $\approx 14\times$ the maximum 86 mL bolus
Actuation	Brushless DC gear-motor with lead-screw compression plate against a rigid backplate, mechanically analogous to a hand squeezing a manual resuscitator bag
Actuation logic	Two roles: (a) reservoir leveling, holding bladder charge between 30% and 80% via inlet throttling; (b) positive bolus assist at settings 4–6, where the plate contributes displacement to the bolus front so the 500 ms window completes without drawing reservoir pressure below 90 kPa gauge
Cycle rate	Breath-triggered primary, 6–40 BPM tracking; Metronome Mode 6–40 cycles/min clinician-selectable
Directional control	Silicone duckbill one-way valves at bladder inlet (from PSA reservoir) and outlet (to delivery line), cracking pressure <0.5 cmH ₂ O; these block rebreathing of exhaled gas and backflow into the PSA reservoir, satisfying checklist item 28
Internal working pressure	2–5 cmH ₂ O at the delivery interface; an open, unsealed low-pressure system by design, not a sealed ventilator circuit
Actuator electrical draw	3–7 W duty-cycled with mode and setting; average 3.5 W at pulse 2
Module mass	0.38 kg including housing

Housing	Glass-filled PA12 shell shaping plate travel and protecting the film from puncture and abrasion; guard-vent pattern keeps the bladder chamber at ambient pressure
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Why a bladder at all, when commercial POCs pulse straight from a valve? Three engineering reasons. First, the buffer decouples the PSA cycle from breath timing, so the compressor runs at its efficiency optimum rather than chasing 40 ms trigger events. Second, the bladder's compliance absorbs the proportional valve's opening transient, delivering a rounded bolus front that users perceive as quieter and gentler than a hard valve snap. Third, at settings 4–6 the plate assist extends the effective bolus window to 500 ms at full volume, which a valve-only architecture of this reservoir size sustains only by oversizing the PSA stage. The cost is 0.38 kg and one annual service item; the Vita Splendida pays it deliberately.

7. Patient Interface, Humidification, and Flow Safety

7.1 Cannula and Mask Interfaces

Table 7-1. Patient interface options

Interface	Specification and use
Standard nasal cannula	Dual-prong, soft-touch PVC-free, 6 mm barbed connector; 2.1 m oxygen-grade tubing; the default interface on all pulse settings and continuous ≤ 2 LPM; weekly-to-monthly replacement as consumable
High-flow nasal cannula	Larger-bore prongs for continuous 2.5–3.0 LPM and Boost; reduces jet velocity and mucosal drying at high flow
Open face mask (OxyMask-class)	Open-diffuser mask for committed mouth breathers on continuous settings; captures oral and nasal flow together; supported as a first-class interface with its own setup screen
Pendant conserving cannula (Oxymizer-class)	Reservoir pendant stores a small oxygen reserve between breaths, extending effective delivery at a given setting; mates to the standard outlet barb

7.2 Humidification Subsystem

Checklist item 8 names nasal drying and mucosal damage as a primary comfort failure of oxygen therapy, concentrated at higher flows. The Vita Splendida integrates humidification into the device rather than bolting on a bottle.

Table 7-2. Humidification subsystem

Element	Specification
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Integrated membrane passover module	Hollow-fiber water-vapor-transfer (Nafion-class) cartridge in the delivery path; 60 mL refillable reservoir; passive, zero power; transfers vapor without bubbling aerosol; raises delivered humidity to $\approx 60\text{--}80\%$ RH at 20 °C reference; active on pulse settings ≥ 3 and all continuous settings
HME cassette (option)	Inline heat-and-moisture-exchange cassette at the cannula for pulse settings 1–3; recaptures exhaled humidity; 30-day consumable
External bubble humidifier (option)	Standard barbed bubble bottle supported on Boost mode and Config B; the device detects back-pressure signature of a bottle and flags it on screen, because bubbling adds $\approx 2\text{--}3$ cmH ₂ O of head that the relief budget accounts for
Reservoir servicing	Top fill port, distilled water; level window plus capacitive level sense with low-water low-priority advisory; reservoir and cartridge are top-rack dishwasher-safe
Physics note	Bubble humidifiers perform poorly under pulsatile flow because gas spends milliseconds in the bottle per bolus; membrane vapor transfer performs identically under pulse and continuous regimes, which is why the Vita Splendida integrates the membrane approach rather than inheriting the bottle habit from stationary concentrators

7.3 Flow Limiting and Pressure Safety

This subsystem exists so no single fault raises delivered flow or pressure past safe cannula limits, satisfying checklist items 25–27 with two independent layers plus a third behavioral layer.

Table 7-3. Flow limiting and pressure safety

Layer	Specification
Primary (electronic)	Proportional solenoid valve under closed-loop mass-flow control; MCU-commanded range 0–4.0 LPM continuous-equivalent; per-bolus volume capped in firmware per setting
Secondary (mechanical failsafe)	Fixed calibrated sapphire orifice in series, sized so gas-path flow never exceeds 6 LPM regardless of upstream pressure or any electronics state; a stuck-open valve, a crashed MCU, and a welded driver all fail into this ceiling
Pressure relief	Spring-loaded pop-off at 5 cmH ₂ O vents to atmosphere on downstream occlusion (kinked tubing, pinched cannula); occlusion detection fires the medium-priority alarm within 3 breaths via outlet-pressure signature
Behavioral layer	Independent safety MCU (Section 8.1) holds direct authority over a latching shut-off valve upstream of the proportional valve; any excursion outside Section 9 limits latches delivery closed and raises the high-priority alarm

Interface pressure class	2–5 cmH2O at the patient interface, open and unsealed; the design excludes sealed positive-pressure ventilation, and the device is not, and never operates as, a ventilator
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8. Control Electronics, User Interface, and Connectivity

8.1 Computing Hardware

Table 8-1. Computing and sensing hardware

Element	Specification
Main MCU	ARM Cortex-M33 class, 160 MHz, hardware FPU, 2 MB flash (STM32U5-class); runs trigger detection, delivery control, compressor loop, UI, and logging
Safety MCU	Independent Cortex-M0+ supervisor with its own clock, watchdog, and direct authority over the latching shut-off valve and alarm transducer; implements the Section 9 fault ladder in hardware-separate firmware per IEC 60601 defensive architecture
Connectivity SoC	Bluetooth LE 5.2 (nRF52-class) serving the remote fob and mobile app; physically separated antenna keep-out; pairing by physical button press, no default keys
Gas-path sensors	Ultrasonic O2 concentration sensor (product reservoir); thermal mass-flow sensor (reservoir outlet); differential pressure + thermistor breath sensors (cannula manifold); bladder pressure transducer; outlet mass-flow sensor
Environmental and motion sensors	Four NTC channels (compressor shell, sieve manifold, each battery bay); barometric altimeter for altitude compensation; 3-axis IMU for activity detection, fall detection, and carry-state awareness
Power management	Per-pack fuel gauge and protector (TI BQ40-class); ideal-diode OR-ing across bays and bridge; USB-C PD 3.1 controller negotiating to 28 V/5 A
Data storage	8 MB onboard flash ring buffer: 30 days of breath rate, delivered volume, concentration, alarms, battery cycles at 1 min resolution; no cloud dependency for core logging

8.2 Display and On-Device Controls

- **Display:** 2.8 in transfective sunlight-readable LCD, 240 × 320, readable in direct sun without backlight and at night with adjustable backlight; primary numerals (mode, flow setting) render at 18 pt equivalent glyph height for low-vision users.
- **At-a-glance ring:** tricolor LED ring around the power button: green (operating in spec), amber (advisory active), red (alarm), visible at 5 m and in peripheral vision.
- **Controls:** three buttons: power (long press), flow up, flow down; mode change requires a deliberate two-button chord, preventing pocket changes; every change confirms by voice

and display.

- **Voice prompts:** six languages at launch; announce mode changes, battery state, and alarms; volume adjustable or muted (alarms excepted, per IEC 60601-1-8).
- **Battery presentation:** per-pack percentage plus system time-remaining estimate at the current setting, computed from a rolling 5-minute power average.

8.3 Remote Fob and Mobile App

Checklist items 20 and 21 demand adjustment while worn, without unpacking the device. The Vita Splendida answers with a dedicated fob plus an app.

Table 8-2. Remote fob and app

Element	Specification
Remote fob	Four buttons: flow up, flow down, mode, alarm silence; haptic click confirms acceptance by the device, not just button press; tricolor status LED mirrors the device ring; 35 × 55 × 8 mm, 18 g; CR2032 cell, ≈12-month life; IP22; 10 m open-air range; clips to clothing or the cannula strap
Mobile app	iOS and Android; mirrors all fob functions plus battery percent per pack, runtime estimate, 30-day usage and saturation-session logs, filter and sieve-life gauges, firmware update over BLE; operates fully offline after pairing
Caregiver read-only link	Optional second pairing gives a caregiver read-only status (no control), supporting remote check-ins on an elderly user without creating a remote-control hazard

8.4 Firmware and Data Logging

Firmware separates into three images with independent update paths: main control (IEC 62304 Class B), safety supervisor (Class C logic, minimal, audited), and connectivity. Updates arrive over BLE through the app or over USB-C at service; the safety image updates only through USB-C with dual-bank rollback. Logs persist 30 days on-device and export as CSV for clinical review. The device functions fully with no account, no cloud, and no subscription; connectivity is convenience, never a dependency, which matters for a device whose users skew older and whose failures are measured in breaths.

9. Alarm and Safety System

The alarm system implements IEC 60601-1-8 three-priority annunciation: high priority (immediate patient-safety risk), medium priority (prompt awareness), and low priority (awareness, no urgency). Audible annunciation runs ≥65 dBA at 1 m for high priority, ≥60 dBA for medium, with distinct temporal patterns per priority; visual annunciation uses the display and the tricolor ring; the fob vibrates on high and medium priority. The safety MCU

owns the alarm transducer directly, so a main-firmware crash degrades the system into a loud safe state rather than a silent one.

Table 9-1. Alarm conditions, priorities, and device behavior

Condition	Priority	Threshold / trigger	Behavior
Low delivered O2 concentration	High	Trend toward 82% (alarm activates before concentration reaches 82%, per ISO 80601-2-69)	Compressor speed-up attempt; latching advisory to switch to backup supply; logged
No flow at outlet	High	Zero outlet flow >30 s in any delivery mode	Checks occlusion vs. disconnect signature; voice prompt plus display graphic identifies which
Occlusion (sustained)	High	Outlet pressure signature at relief threshold across 3 breaths	Pop-off vents at 5 cmH2O; prompt to unkink tubing; auto-clears on resolution
System fault	High	Safety MCU detects sensor disagreement, valve failure, or main-MCU watchdog expiry	Latching shut-off valve closes; continuous tone; device demands restart
Battery critical	High	<5 min energy at current draw across all sources	Continuous tone; voice countdown; graceful shutdown with 60 s delivery reserve
Over-temperature	Medium then High	Compressor shell >75 °C or any pack >45 °C	Derate compressor 20% (medium); shutdown at +5 °C further (high)
Concentration advisory	Medium	87% for >60 s	Sieve-life check prompt; logged for service
High breath rate	Medium	>40 BPM sustained 60 s	Advisory only; delivery continues per Section 5.7
No breath detected	Medium	15 s without trigger in pulse mode	Metronome fallback (or continuous in Sleep Mode); cannula-check graphic
Cannula disconnect	Medium	Loss of outlet back-pressure signature	Voice plus graphic prompt; delivery continues for safety
Low battery	Medium	15 min at current draw	Voice and display; fob LED amber
External power lost	Medium	AC/DC removed while running	Seamless battery take-over; advisory tone per home-care convention

Filter service due	Low	Intake restriction trend or 30-day timer	Display badge; rinse instructions on screen
Sieve cartridge life	Low	Concentration margin trend plus 12-month timer	Replacement prompt; cartridges user-replaceable tool-free
Humidifier low water	Low	Capacitive level sense	Display badge; refill prompt

Safety architecture summary. Dual flow limiting (electronic plus mechanical orifice), spring pop-off at 5 cmH2O, duckbill one-way valves, per-pack battery protection plus system-level protection, an independent safety MCU with latching shut-off authority, and a three-tier alarm suite with a 30-day tamper-proof log. Every layer operates independently of the one above it; no two share a single point of failure.

10. Power Subsystem

10.1 Battery Architecture

The Vita Splendida-6 taught one lesson above all: a single fixed 400 Wh pack sized for 6 hours at 63 W weighs 2.7–3.1 kg and chains the user to a wall for as long as it ran. The Vita Splendida inverts that architecture into three tiers.

Table 10-1. Power architecture

Element	Specification
Hot-swap packs (×2)	98 Wh each, 4s2p 21700 NMC (14.4 V, 6.8 Ah), ≈205 Wh/kg at pack level, 0.46 kg each; keyed bay with ejector latch and recessed contacts; each pack carries its own fuel gauge and protector; 98 Wh sits under the 100 Wh threshold for unrestricted passenger carriage of installed and spare lithium batteries on commercial aircraft
Internal bridge battery	40 Wh, non-removable, standard equipment; sustains the full load through any pack swap of unlimited duration; alone delivers ≈1.2 h at pulse 2; satisfies checklist item 12 (built-in battery as standard)
Hot-swap behavior	Ideal-diode OR-ing across both bays and the bridge; the user ejects either pack at any time and inserts a charged pack; delivery, alarms, and settings persist without interruption or reboot; the display confirms charge acceptance within 3 s
Charge acceptance	Per-pack 2 A–4 A CCCV under pack-local control; both packs plus bridge charge simultaneously from a 140 W source
Cycle life	≥500 cycles to 80% capacity per pack; pack-local logging drives a health readout in the app; end-of-life packs return through the recycling program (Section 15)

Chemistry choice

NMC 21700 over LiFePO4: pack-level ≈ 205 Wh/kg versus ≈ 125 – 140 Wh/kg for LFP, the difference between 0.46 kg and 0.75 kg per 98 Wh; thermal runaway risk is managed by pack-local protection, cell spacing, and the 45 °C thermal ladder, an acceptable trade for a worn device where mass is the dominant patient-outcome variable

10.2 Runtime Budget

Runtime figures derive from measured-class average draws (Table 10-2) against 180 Wh of usable dual-pack energy (196 Wh nominal $\times$ 0.92 conversion and end-of-life derate). Figures quote at 20 BPM in pulse modes; runtime shortens roughly 15% at 35 BPM.

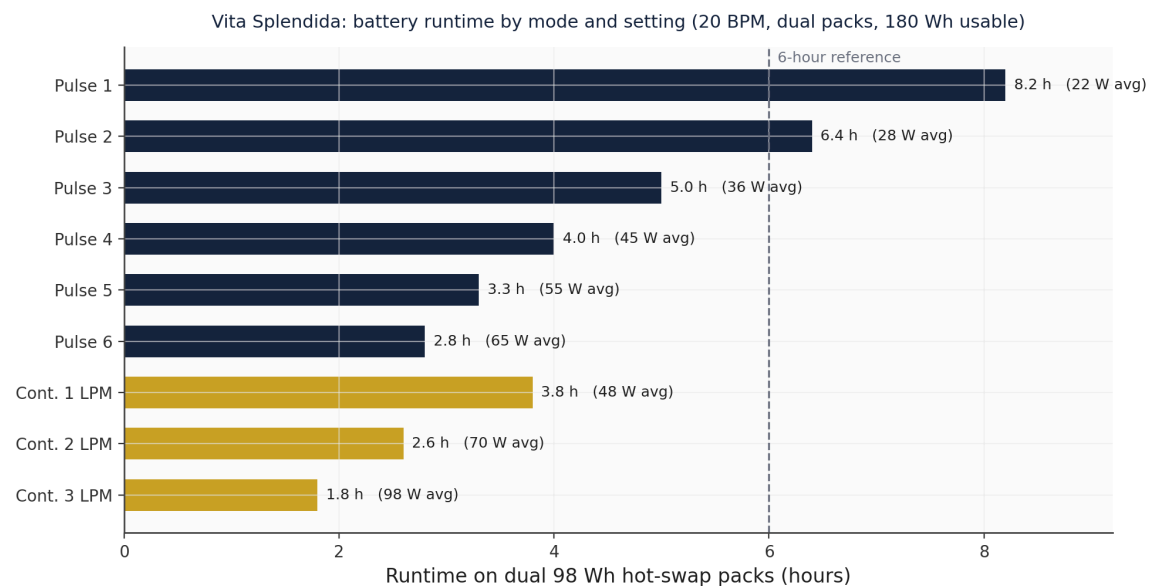


Figure 10-1. Dual-pack runtime by mode and setting. The 6-hour checklist target (dashed line) is met at pulse settings 1–2 with margin, and hot-swap extends any figure indefinitely.

Table 10-2. Average draw and runtime by mode and setting

Mode / setting	Average system draw	Dual-pack runtime (196 Wh)	Single-pack runtime (98 Wh)
Pulse 1	22 W	8.2 h	4.1 h
Pulse 2	28 W	6.4 h	3.2 h
Pulse 3	36 W	5.0 h	2.5 h
Pulse 4	45 W	4.0 h	2.0 h
Pulse 5	55 W	3.3 h	1.6 h
Pulse 6	65 W	2.8 h	1.4 h
Continuous 1.0 LPM	48 W	3.8 h	1.9 h
Continuous 2.0 LPM	70 W	2.6 h	1.3 h

Continuous 3.0 LPM	98 W	1.8 h	0.9 h
Boost 4.0 LPM	140 W	30-min excursions (external-power mode by design)	Not recommended

10.3 Charging and External Power

Table 10-3. Charging options

Source	Specification
AC wall supply	140 W USB-C PD 3.1 GaN adapter (100–240 V, 50/60 Hz, auto-sensing); runs the device at any setting including Boost and charges both packs plus bridge simultaneously; dual packs recharge 0 → 100% in ≈1.8 h, 0 → 80% in ≈55 min
Car DC adapter	100 W, 11–16 V input; runs the device and charges simultaneously up to pulse 4 or 2.0 LPM continuous; at higher settings it extends rather than fully offsets battery draw, displayed honestly on screen; satisfies checklist item 14
Desktop dual-bay charger (accessory)	Charges two spare packs off-device in ≈1.5 h, enabling a two-pack rotation: two in the device, two on the charger, effectively unlimited away-from-outlet operation
USB-C PD source flexibility	Any PD 3.0/3.1 source ≥65 W runs the device at pulse ≤3 and charges at reduced rate; a 65 W laptop-class adapter or a 25,000 mAh PD power bank becomes a legitimate travel power strategy

11. Thermal and Acoustic Management

The Vita Splendida-6 identified the core conflict: acoustic damping wants to seal the compressor in foam, while cooling wants open airflow across it. The Vita Splendida resolves the conflict with a ducted acoustic plenum rather than choosing between the two. Intake and exhaust air travel separate labyrinth paths lined with open-cell acoustic foam; sound reflects and attenuates through the labyrinth while air passes freely. A thermostatically controlled 40 mm fan (1–2 W) assists convection only above 60 W compressor duty, so low-setting operation stays fan-silent.

Table 11-1. Thermal budget at the two design extremes (20 °C ambient)

Heat source	Pulse 2 (28 W system)	Cont. 3 LPM (98 W system)	Mitigation
Compressor (motor + compression loss)	≈17 W	≈78 W	Finned aluminum shell in the ducted airflow path; variable speed keeps duty matched to demand

Sieve-bed cycling	≈2 W	≈6 W	Conducted into the same plenum airflow
Battery I ² R (both bays + bridge)	≈2 W	≈6 W	Separate battery thermal zone with membrane vents; 45 °C derate ladder (Section 9)
Electronics, valves, sensors	≈3 W	≈4 W	Chassis conduction; negligible
Cooling fan	0 W (off)	1.8 W	Runs only above 60 W duty; speed-ramped

Thermal limits: compressor shell 75 °C derate / 80 °C shutdown; battery zone 45 °C derate / 50 °C shutdown; exhaust air at the chassis vent ≤48 °C at 3 LPM continuous, and vent geometry directs it away from the wearer's body and the cannula tubing.

Table 11-2. Acoustic performance targets (A-weighted, 1 m, free field, device in backpack carry)

Operating point	Vita Splendida target	Commercial reference
Pulse 1–2 (fan off)	40 dBA	Rove 6: 37–39 dBA; P2-E7: 39 dBA
Pulse 4	43 dBA	FreeStyle Comfort: 43 dBA at setting 2
Pulse 6	46 dBA	P2-E7: 48 dBA at setting 7
Continuous 2.0 LPM	48 dBA	SimplyGo: 43 dBA at pulse 2; Oxlife: 40 dBA at pulse 2
Continuous 3.0 LPM / Boost	52 dBA	Stationary 5 LPM class: 45–55 dBA

Honest acoustics. The scroll compressor's valve-less, pulsation-free character is what makes 40 dBA credible at pulse 2: the dominant residual noise is intake rush and motor tone, both of which the labyrinth plenum attenuates well. At 3 LPM continuous the fan and 98 W of compression push the floor to roughly 52 dBA, which resembles a quiet refrigerator hum. The Vita Splendida never reaches the 55–65 dBA of the Vita Splendida-6's unenclosed design, and it also never claims the fictional 45 dBA-at-all-settings figure that marketing sheets favor. Alarm transducers bypass the plenum and announce at ≥65 dBA as Section 9 specifies.

12. Enclosure, Harness, and Wearability

Table 12-1. Enclosure and wearability

Parameter	Specification
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Dimensions	26.5 × 18.5 × 9.0 cm (L × W × H), ≈4.4 L envelope; crescent-curved shell conforms to the lumbar curve in backpack mode and the hip in cross-body mode, eliminating the walking bounce boxy units exhibit
Worn mass	3.50 kg dual-pack; 3.04 kg single-pack; mass distribution centers low and within 4 cm of the spine in backpack mode, minimizing added respiratory work (checklist items 9 and 15)
Shell	Glass-filled PA12, 2.5 mm wall, rib-stiffened; survives a 1 m drop onto hardwood per IEC 60601-1-11 home-environment mechanical regime; UV-stabilized
Ingress protection	IP22: protected against finger-scale objects and dripping water at 15° tilt; expanded-PTFE membrane vents pass air, block liquid; conformal-coated PCBAs; sweat and light rain tolerated (checklist item 17)
Skin-contact materials	Strap padding in closed-cell TPU foam with recycled-polyester knit cover; all skin-contact and gas-path materials evaluated per ISO 10993-1/-5/-10 and ISO 18562; latex-free throughout (checklist item 16)
Carry systems	Backpack (dual padded straps, sternum and waist stabilizers), cross-body messenger strap, shoulder bag, and belt-clip plate for single-pack configuration; tool-free swap between systems in <30 s (checklist item 13)
Controls while worn	On-device panel faces forward in cross-body carry; remote fob and app operate in all carry modes (checklist items 20–21)
Environmental envelope	Operating 5–40 °C, 15–90% RH non-condensing, 0–3,048 m standard (advisory-limited to 4,000 m); storage –20 to 60 °C, 5–90% RH

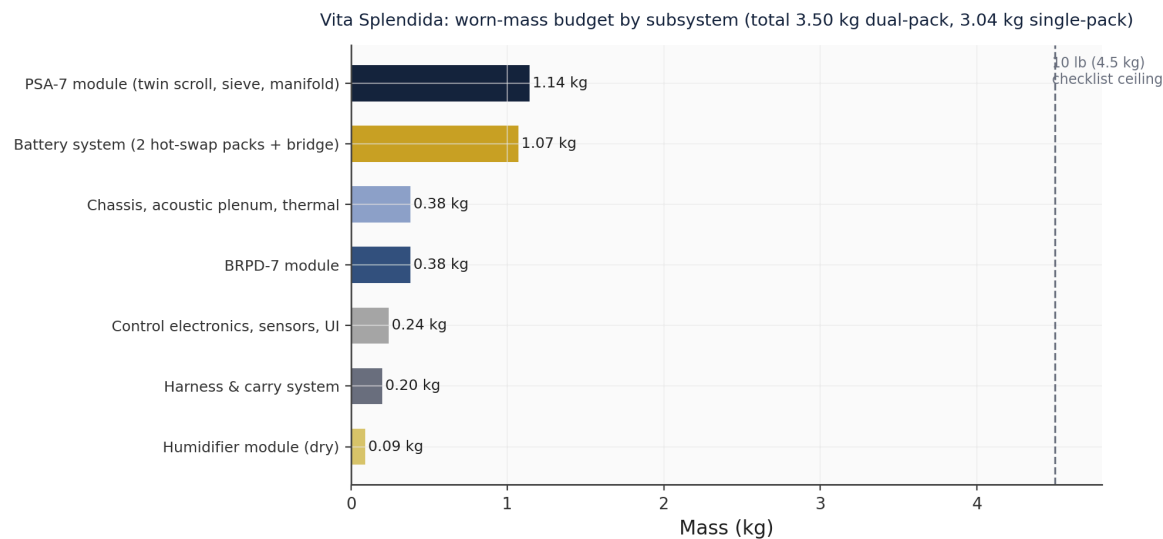


Figure 12-1. Worn-mass budget by subsystem. The two hot-swap packs plus bridge total 1.07 kg, versus 2.7–3.1 kg for the Vita Splendida-6's single fixed pack; total worn mass lands 1.0 kg under the checklist ceiling.

13. Configuration B: Tethered Base Station

Configuration A (Sections 3–12) carries everything on the body. Configuration B retains the Vita Splendida-6's honest second option for users whose need profile centers on high continuous flow at home with mobility inside a hose radius.

Table 13-1. Configuration B base-station arrangement

Element	Specification
Base station	Mains-powered stationary unit on a wheeled cart: larger twin-scroll (≈ 45 L/min free air delivery), 5.0 LPM continuous at 90% $\pm 3\%$ as a standard rating, not an exception; ≈ 45 dBA at 2 LPM in a larger, heavier acoustic enclosure where mass buys quiet
Supply link	2–3 m oxygen-grade supply hose with quick-connects both ends, carrying enriched gas at ≈ 140 kPa gauge to the worn unit; breakaway coupling vents safely on snag
Worn unit in Config B	The Vita Splendida device itself, undocked from its PSA stage conceptually but not physically: the simplest Config B keeps the whole Vita Splendida on the user and feeds it base-station gas through an inlet port, with the onboard compressor idled; alternatively a dedicated BRPD-vest (BRPD-7, flow limiting, electronics, one 98 Wh pack) at ≈ 1.4 kg worn
Runtime	Effectively unlimited on mains; the worn pack covers untethered excursions (bathroom, garden) at full specification, then re-docks
Best fit	Users prescribed 4–5 LPM continuous at home, users who sleep on continuous flow, and facilities standardizing on one device ecosystem across ambulatory and stationary use

Configuration B exists because physics does not negotiate: sustained 5 LPM at 90% purity wants roughly 43 L/min of compressed air and 180–220 W of continuous power. A wearable answers that demand for minutes; a base station answers it forever. The Vita Splendida architecture treats both answers as one product family rather than pretending the question away.

14. Performance Specification Summary

Table 14-1. Vita Splendida master specification, Configuration A

Specification	Value
Delivered O2 concentration	90% $\pm 3\%$ at all settings (93% nominal ≤ 2 LPM continuous); alarm activates before 82% per ISO 80601-2-69
Pulse settings	1–6; bolus 15–86 mL at 20 BPM reference ($\pm 15\%$ or 4 mL); 1,500 mL/min maximum oxygen output

Continuous flow	0.5–3.0 LPM in 0.5 steps on battery; Boost 3.5–4.0 LPM on external power
Breath trigger	–0.08 cmH ₂ O standard, adjustable –0.05 to –0.15; <40 ms latency; bolus inside first 250 ms of inspiration
Breath-rate tracking	6–40 BPM adaptive; 50 BPM hardware maximum; Metronome fallback 6–40 cycles/min
Sleep Mode	–0.05 cmH ₂ O trigger; 15 s no-breath fallback to 1.5 LPM continuous with advisory
Bladder (BRPD-7)	1.2 L LDPE/EVOH; duckbill one-way valves; user-replaceable annually
Delivery pressure	2–5 cmH ₂ O open interface; pop-off relief at 5 cmH ₂ O; fixed orifice hard-caps gas path at 6 LPM
Humidification	Integrated membrane passover module (60 mL); HME cassette option; bubble-bottle support on Boost
Battery	2 × 98 Wh hot-swap NMC 21700 packs + 40 Wh internal bridge; 180 Wh usable dual
Runtime (dual-pack)	8.2 h pulse 1; 6.4 h pulse 2; 5.0 h pulse 3; 2.8 h pulse 6; 3.8 h cont. 1 LPM; 2.6 h cont. 2 LPM; 1.8 h cont. 3 LPM
Recharge	≈1.8 h dual packs on 140 W USB-C PD 3.1; 80% in ≈55 min; 100 W car DC adapter
Worn mass	3.50 kg dual-pack; 3.04 kg single-pack
Dimensions	26.5 × 18.5 × 9.0 cm
Acoustics	40 dBA pulse 2; 46 dBA pulse 6; 48 dBA cont. 2 LPM; 52 dBA cont. 3 LPM / Boost; alarms ≥65 dBA
Interface	2.8 in transfective LCD; three buttons; voice prompts, six languages; BLE remote fob; iOS/Android app; caregiver read-only link
Alarms	15 conditions, IEC 60601-1-8 three-priority; independent safety MCU with latching shut-off
Environment	5–40 °C; 15–90% RH non-condensing; 0–3,048 m (advisory to 4,000 m); IP22; Class II equipment, Type BF applied part
Service life	8-year device target; compressor 15,000 h; sieve cartridges 12 months; packs ≥500 cycles to 80%
Warranty	3-year device and compressor (5-year option); 2-year sieve cartridges; 1-year batteries and accessories; 48-h advance exchange
Target price	\$995 direct channel; \$1,295 retail bundle; costed BOM \$507 at 10,000 units/year (Section 16)

15. Reliability, Maintenance, and Warranty

15.1 Scheduled Maintenance

Table 15-1. Maintenance schedule (all user-level unless noted)

Interval	Task	Consumable / tooling
Weekly	Rinse intake foam pre-filter; air dry; wipe shell with damp cloth; check cannula for stiffness	Water only; no tools
Monthly	Descale humidifier reservoir (1:3 white vinegar soak, rinse); inspect tubing; check pre-filter replacement need	Household vinegar
Every 3 months	Replace fine particulate intake filter (quarter-turn bayonet)	Filter kit, \$8
Every 12 months	Replace sieve cartridges (pair; tool-free); replace BRPD bladder; ultrasonic-sensor cross-check at service center or by mail-in kit	Cartridge pair \$89; bladder \$19
As indicated	Cannula (2–4 weeks typical), HME cassette (30 days), humidifier cartridge (6 months), battery packs (~500 cycles)	Cannula \$4; HME \$12; cartridge \$24; pack \$129

The device tracks each interval and displays remaining life as a percentage for filters, sieve, and packs, because a maintenance program the user cannot see is a maintenance program that does not happen.

15.2 Reliability Design Targets

- **Compressor:** 15,000 h design life (scroll air-end, no valves, no wearing contacts in the compression chamber); corresponds to 5+ years of 8 h/day duty.
- **Device service life:** 8 years target, matching the best published figure in the commercial class.
- **Drop and shock:** 1 m drop, all six faces, onto hardwood; 25,000-cycle strap-flex; 100,000-cycle pack-eject and cartridge-bayonet endurance.
- **Alarm-system availability:** the safety MCU and transducer operate on bridge-battery reserve even with both packs removed, so the device never goes silently dead.
- **MTBF target:** $\geq 20,000$ h electronics; field-return target <3% in year one, with the advance-exchange program absorbing the remainder.

15.3 Warranty and Support Program

Table 15-2. Warranty and support

Element	Terms
Device and compressor	3 years standard, parts and labor; optional extension to 5 years at \$149
Sieve cartridges	2 years (12-month replacement cycle covered one-for-one)
Batteries and accessories	1 year
Advance exchange	48-hour replacement unit shipped before return of the failed unit, so therapy never pauses for logistics
Support	24/7 phone and chat; remote diagnostics via app log export; loaner program during service
Trial and returns	30-day satisfaction return; recycled-pack return credit \$15 per pack

16. Cost Model and Target Pricing

Checklist item 30 sets the target: roughly \$1,000, against a commercial field listing at \$2,795–\$3,995. Table 16-1 builds the costed bill of materials at 10,000 units per year with contract manufacturing, the volume at which a direct-to-consumer channel carries a device of this class credibly. Figures are engineering estimates, stated with their assumptions, not quotations.

Table 16-1. Costed bill of materials and build cost (USD, 10,000 units/year)

Item	Basis	Unit cost
Twin-scroll compressor, oil-free, custom air-end	Volume contract, Air Squared-class OEM pricing at 10k/yr	\$74
Sieve cartridges (2) incl. LiLSX charge ≈280 g	Zeolite ≈\$12 + housings, valves seats, dust filters	\$26
PSA manifold and solenoid valve set	Machined manifold + 6 pilot valves	\$34
BRPD-7 module	Film bladder, gear-motor, lead screw, duckbills, shell	\$27
Main PCBA + safety MCU board	Cortex-M33 + M0+, power stages	\$50
Sensors (ultrasonic O2, mass flow ×2, pressure/thermal breath, NTCs, IMU, baro)	Ultrasonic O2 sensor dominates at ≈\$22	\$43
Display + controls + alarm transducer	2.8 in transfective LCD, membrane buttons	\$14
Battery system (2 × 98 Wh packs + 40 Wh bridge)	16 × 21700 cells ≈\$51, BMS and housings	\$84

Chassis, acoustic plenum, thermal hardware	PA12 shell, foam, heatsink, fan	\$31
Harness and carry systems	Backpack + cross-body + shoulder bag	\$18
Humidifier module	Membrane cartridge + reservoir	\$9
Box accessories (140 W GaN AC supply, car DC cable, remote fob, cannulas, filters, documentation)	AC supply \$22, DC cable \$11, fob \$13, consumables \$6	\$52
Assembly, calibration, test	35 min build + 100% flow/purity/alarm test	\$38
Packaging	Recyclable molded pulp + print	\$7
Build cost subtotal		\$507
Warranty reserve (4%)	Actuarial on 3% year-one field-return target	\$20
Landed cost		\$527

Pricing logic: a \$995 direct-channel price yields $\approx 47\%$ gross margin, funding the 24/7 support line, the advance-exchange pool, certification amortization (Section 17), and the consumables ecosystem. A \$1,295 retail bundle adds a third battery pack and the desktop dual-bay charger. Three honest dependencies attach to the \$995 figure: 10,000+ units/year of compressor volume, a direct-sales channel without DME-middleman margin stacking, and reimbursement-free cash sale. Below 3,000 units/year the landed cost rises roughly 35% and the price migrates toward \$1,395; Section 18 keeps this visible rather than buried.

17. Regulatory Pathway and Standards

The Vita Splendida is a conceptual design study, and this section maps the full gauntlet between this document and lawful human use. Nothing here is optional in practice, and the schedule estimate below assumes a funded, competent regulatory program.

Table 17-1. Standards and regulatory map

Instrument	Scope for the Vita Splendida	Phase estimate
FDA 510(k), Class II	Portable oxygen concentrator, 21 CFR 868.5440 (product code CAW); predicate devices exist across the benchmark table (Rove 6, FreeStyle Comfort, SimplyGo)	Pre-submission to clearance: 12–18 months after design freeze
Health Canada MDL	Class II under the Medical Devices Regulations (SOR/98-282); leverages the same technical file	Parallel, +2–3 months

IEC 60601-1	Basic safety and essential performance, general standard	Test campaign: 6–9 months on frozen design
IEC 60601-1-2	Electromagnetic compatibility, home and transport environments	
IEC 60601-1-8	Alarm-system verification against Section 9	
IEC 60601-1-6 / IEC 62366-1	Usability engineering; the three-button interface, voice prompts, and fob are the usability-mitigation set	
IEC 60601-1-11	Home healthcare environment: IP22, drop regime, lay-user documentation	With test campaign
ISO 80601-2-69	Oxygen concentrator particular requirements: 90% ±3% across settings, concentration monitor with alarm before 82%, outlet-pressure limits	
ISO 18562-1/-2/-3	Gas-pathway biocompatibility: particulates, VOCs, leachables across the full flow path including bladder film and humidifier cartridge	
ISO 10993-1/-5/-10	Skin-contact biocompatibility for straps and touch points	
IEC 62304 + ISO 14971	Software lifecycle (Class B main / Class C logic safety supervisor) and risk management file; the dual-MCU architecture is the core risk control	Continuous through development
IEC 62133-2, UN 38.3, UL 2054	Battery pack safety and transport testing; UN 38.3 test summary ships with every pack	3–4 months, parallel
FAA acceptance + IATA provisions	POC acceptance criteria for carriage and use aboard aircraft (labeling, emissions per RTCA DO-160 practice); 98 Wh packs sit under the 100 Wh unrestricted threshold, with airline approval available to 160 Wh for up to two spares	Documentation exercise post-EMC

Bench validation precedes all of it: PSA purity and flow mapping across the full mode matrix and environmental envelope, trigger-latency characterization on a breathing simulator across 6–50 BPM, bolus-window verification by fast O₂ analyzer at the nares, accelerated compressor life testing, and alarm-fault injection against every Table 9-1 row. A realistic total from design freeze to first lawful sale: 18–24 months and \$2.5–4M in verification, testing, and regulatory cost, amortized in the Section 16 margin.

18. Honest Limits and Open Trade-Offs

In the tradition the Vita Splendida-6 established, this section collects every limit a marketing department would prefer to delete. Each is real, each has an engineering reason, and each appears here so the document's other claims carry weight.

- **Sustained continuous flow above 3 LPM does not fit in a 3.5 kg wearable.** At roughly 8.5 volumes of compressed air per volume of 90% product, 4 LPM wants ≈ 150 W continuously. On 180 Wh of usable battery that is a 75-minute device. The Vita Splendida therefore gates Boost to external power for sustained use and points 5 LPM users to Configuration B. Any competitor claiming long-runtime 5 LPM continuous from a sub-4 kg battery device is quoting pulse numbers as if they were continuous.
- **Pulse settings are not liters per minute.** Setting 6 (86 mL boluses) serves many users prescribed 4–5 LPM-equivalent ambulatory dosing, but equivalence is clinical, titrated by oximetry, not arithmetic. The manual states this in the standard industry wording, and the display never renders pulse settings as liter figures.
- **Mouth breathing defeats some pulse conservation, always.** Dual sensing and Oral-Adaptive logic recover much of it, and the mask interface on continuous mode answers the rest, but a pure mouth breather on pulse mode wastes gas no sensor rescues. The correct prescription answer is continuous or mask, and the Vita Splendida supports both rather than denying the problem.
- **The Vita Splendida is 1.3 kg heavier than the lightest pulse-only devices.** A Rove 6 carries 2.2 kg because it omits continuous flow, humidification, dual hot-swap power, and the BRPD stage. A pulse-only user at setting ≤ 3 is better served by that lighter class. The Vita Splendida's mass is the price of its mode coverage, stated plainly.
- **40 dBA at pulse 2 is quiet, not silent.** In a library-quiet room the intake rush remains audible within a few meters. 52 dBA at 3 LPM continuous resembles a refrigerator. No 98 W compressor runs quieter at this mass, and claims to the contrary elsewhere deserve skepticism.
- **Sieve life halves in humid climates.** Zeolite adsorbs water preferentially. The 12-month cartridge interval assumes temperate conditions; the concentration-margin trend alarm exists precisely because tropical users approach 6–8 months. Cartridges are user-replaceable at \$89 a pair to keep that event cheap.
- **The \$995 price is volume-dependent.** Below 3,000 units/year the landed cost rises $\approx 35\%$ and honest pricing moves to $\approx \$1,395$. Affordability at scale is a business-model commitment, not a component miracle.
- **The BRPD bladder is a service item.** PE film eventually flex-fatigues. Annual \$19 replacement is designed in and tool-free; treating it as permanent would be the kind of quiet unreliability this document exists to avoid.
- **This is a design study, not a device.** Every figure in this document is an engineering target grounded in measured commercial performance and published component data, and none of it substitutes for the bench validation and regulatory clearance Section 17 maps. No human breathes from this device until that gauntlet completes.

19. Revision History

Table 19-1. Document lineage

Revision	Date	Substance
Vita Splendida-6 Rev. A	Prior	Initial conceptual architecture: PSA plus PE-film bellows, fixed-rate pacing, 45 dBA acoustic target, no thermal section
Vita Splendida-6 Rev. B	Prior	External design review corrections: acoustics revised to 55–65 dBA, thermal management added, BRPD renamed honestly, tethered Configuration B introduced, battery weight penalty stated
Vita Splendida Rev. 1.0	July 16, 2026	This document. Re-engineered against the 30-point patient checklist and the 2024–2026 commercial state of the art: breath-triggered delivery, true continuous flow to 3 LPM plus Boost, dual hot-swap power with bridge, integrated humidification, remote fob and app, full IEC 60601-1-8 alarm suite, honest PSA re-dimensioning, costed BOM to a ≈\$1,000 target, warranty and regulatory pathway

Reference Data Sources

Benchmark and component figures in this document derive from the following manufacturer and technical sources, current as of July 2026. Engineering targets specific to the Vita Splendida are the authors' own and appear as such throughout.

1. Inogen, Inc. *Inogen Rove 6 Portable Oxygen Concentrator Specifications and User Manual* (Model IO-501/IS-501), 2023–2024: mass 2.2 kg, six pulse settings to 1,260 mL/min, 39 dBA at setting 2, trigger sensitivity <0.12 cmH₂O, battery durations, \$2,795–\$3,295 list, 3-year warranty.
2. CAIRE Inc. *FreeStyle Comfort Product Brochure and User Manual*: mass 2.3–2.7 kg, five pulse settings to 1,050 mL/min, 39.9–43 dBA at setting 2, battery durations by pack, UltraSense/autoSAT descriptions, 3-year warranty including sieve beds.
3. Rhythm Healthcare. *P2-E6/P2-E7 Portable Oxygen Concentrator User Manuals and Listings*: mass 1.98 kg, settings to 1,400 mL/min, 39–48 dBA, runtime table, IP22 / Class II / Type BF classifications, 5-year warranty, ≈\$3,995 list.
4. Philips Respironics. *SimplyGo User Manual and Specification Sheets*: mass 4.5 kg, continuous 0.5–2.0 LPM plus pulse bolus table (12–72 mL), Sleep Mode behavior with 2 LPM no-breath fallback, 43 dBA, alarm list, power and battery figures.
5. O2 Concepts LLC. *Oxlife Independence Specifications*: continuous 0.5–3.0 LPM plus pulse 8–96 mL, dual-battery hot-swap operation, ≈90 W average at setting 2, 2.5 h at 2 LPM continuous, 40 dBA at pulse 2, 5-year warranty, \$2,895–\$3,195 list.
6. Air Squared, Inc. *Portable Oxygen Concentrator Compressor Solutions; P07H015A-BLDC-C Scroll Compressor*: oil-free scroll lineage for POCs, variable-speed BLDC, low-noise characteristics.
7. Baksh, M.S.A., Kikkinides, E.S., Yang, R.T. "Lithium Type X Zeolite as a Superior Sorbent for Air Separation," *Separation Science and Technology* 27(3), 1992: Li-X N₂/O₂ selectivity ≈7 at 1 atm; >90% O₂ purity at >70% recovery versus <30% for Na-X.
8. WHO/UNICEF. *Oxygen Concentrator Target Product Profile* and *WHO Technical Specifications for Oxygen Concentrators* (2015): ISO 80601-2-69 requirement that the low-concentration alarm activate before

delivered concentration falls below 82%; standards suite (IEC 60601-1, -1-2, -1-6, -1-8, -1-9, -1-11; ISO 13485; ISO 14971).

9. ModELAS wearable artificial lung ovine 1-month study, *ASAIO Journal* (PMC8024407); University of Pittsburgh PAAL program (NIH \$3.4M); University of Maryland artificial pump-lung; Hemovent Mobybox: blood-side extracorporeal lung-assist landscape, Section 2.3.
10. Commercial battery references: 21700 NMC cell energy density 250–300 Wh/kg, pack-level 150–205 Wh/kg; FAA/IATA lithium battery passenger provisions (≤ 100 Wh unrestricted, 101–160 Wh with airline approval, two spares maximum).